

Software Engineer

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Experience

SDE 2

*Remote**VideoVerse**Jan 2026 – present*

- Built SmartThumbnails from scratch — an AI-powered thumbnail selection service that processed **4,500+ clips in its first month with a 99.2% client acceptance rate**, effectively eliminating manual thumbnail curation from broadcast workflows.
- Engineered the service using **FastAPI, PySceneDetect** for scene segmentation, Laplacian sharpness scoring for frame quality filtering, and Google Gemini via **DSPy** for AI-driven final selection
- Automated clip enrichment at scale via an event-driven pipeline, making every clip searchable and tagged the moment it's ingested.
- Added multi-language, EPG metadata, and AI tagging support to the export pipeline, **processing 11,000+ clips monthly** and letting broadcast clients localize content without manual work.
- Automated conflict detection and metadata back-fill across sibling clips, reducing manual editorial effort and **SLA breaches by 25%**.
- Shipped an API that unified a broadcast client's scheduling data into a single queryable source, replacing fragmented manual imports
- Migrated from **Datadog to OpenTelemetry**, giving the team vendor-neutral structured observability and faster production debugging.

SDE 1

*Remote**VideoVerse**Nov 2024 – Jan 2026*

- Built an internal LLM evaluation pipeline using Langfuse and **DSPy Evaluate**, improving model output accuracy **from 68% to 92%**.
- Applied automated prompt optimization using **DSPy's MIPRO/COPRO** optimizers and **GEPA**, lifting task-level accuracy **from 30% to 87%**.
- Built a centralized **AI Gateway** handling auth, request tracking, and auto-retriggers across 5 backend AI services, processing **50,000+ clips and 300+ streams monthly**
- Engineered an LLM-powered video clip ratings pipeline using **DSPy**, handling over **500k clips per month**
- Established monitoring and alerting workflows using Langfuse and OTEL to enforce SLA compliance.
- Co-led migration of entire **AWS infrastructure to Terraform**, improving infrastructure-as-code practices.

Software Engineer Intern

*Remote**VideoVerse**June 2024 – Nov 2024*

- Originally interned at **LetsDive.io** as a Django backend engineer, leading the codebase migration from ***Python* 3.9 to 3.11** continued the role at VideoVerse **following its acquisition of LetsDive.io**.
- Designed and implemented an ETL pipeline with **sub-100ms latency** to optimise data processing.
- Improved an existing thumbnail service by migrating from **SQLite to PostgreSQL**, adding auto-retries, and enhancing face-embedding data collection for better selection accuracy

Projects

Battlefield Portal Library | bfportal.gg | <https://gh.bfportal.gg>

Jan 2022 – present

- Built a Django-based CMS serving **17,000+ users**, with a full frontend built on Jinja templates — currently migrating to **HTMX** for faster, dynamic interactions without a JavaScript framework.
- Containerised with Docker and automated deployments via GitHub Actions.

Gametools Network | gametools.network | github.com/community-network

May 2022 – present

- Developed and maintain an open-source Stats API serving **3 million players**, handling over **2 million daily requests** via Envoy proxy and Cloudflare caching.

Skills

AI / ML: DSPy, MIPRO/COPRO, GEPA, Langfuse, Google Gemini, VertexAI, Prompt Engineering, LLM Evaluation & Optimization

Backend: Python, Go, TypeScript, FastAPI, Django, PostgreSQL, MongoDB, Redis

Infrastructure: AWS, Terraform, Docker, OpenTelemetry, Grafana, Nginx, GitHub Actions, Envoy

Education

MIET

*Meerut, Uttar Pradesh**B.Tech in Computer Science Engineering, Minor in Data Science**Jan 2020 – Jan 2024*